LA HW

Included exercises: 5, 8-9, 14, 16, 53-55

Exercises 1-8

In Exercises 1–8, determine whether each set is orthogonal.

Original Exercise 5

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ -5 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix} \right\}$$

Original Exercise 8

$$\left\{\begin{bmatrix}2\\1\\-1\\1\end{bmatrix},\begin{bmatrix}1\\1\\3\\0\end{bmatrix},\begin{bmatrix}1\\-1\\0\\1\end{bmatrix}\right\}$$

ram–Schmidt process to
t set S by an orthogonal
span, and (b) obtain an
s S.

Exercise 9

$\begin{bmatrix}-1\end{bmatrix} \quad \begin{bmatrix}5\end{bmatrix}$ $\qquad$ $\begin{bmatrix}4\end{bmatrix} \quad \begin{bmatrix}3\end{bmatrix}$

$\begin{bmatrix}0\end{bmatrix} \quad \begin{bmatrix}1\end{bmatrix} \quad \begin{bmatrix}1\end{bmatrix}$ $\qquad$ $\begin{bmatrix}1\end{bmatrix} \quad \begin{bmatrix}1\end{bmatrix} \quad \begin{bmatrix}3\end{bmatrix}$

Original Exercise 9

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ -1 \\ 2 \end{bmatrix} \right\}$$

Original Exercise 14

$$\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \\ 5 \end{bmatrix} \right\}$$

Original Exercise 16

$$\left\{ \begin{bmatrix} -1 \\ 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$

Original Exercise 53

Let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ be an orthogonal subset of $\mathcal{R}^n$. Prove that, for any scalars $c_1, c_2, \ldots, c_k$, the set

Original Exercise 54

$\{c_1\mathbf{v}_1, c_2\mathbf{v}_2, \ldots, c_k\mathbf{v}_k\}$ is also orthogonal.

Suppose that $\mathcal{S}$ is a nonempty *orthogonal* subset of $\mathcal{R}^n$ consisting of nonzero vectors, and suppose that $\mathcal{S}'$ is obtained by applying the Gram–Schmidt process to $\mathcal{S}$.

Original Exercise 55

Prove that $\mathcal{S}' = \mathcal{S}$.

Let $\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_n\}$ be an orthonormal basis for $\mathcal{R}^n$. Prove that, for any vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathcal{R}^n$,

(a) $\mathbf{u} + \mathbf{v} =$
$(\mathbf{u} \bullet \mathbf{w}_1 + \mathbf{v} \bullet \mathbf{w}_1)\mathbf{w}_1 + \cdots + (\mathbf{u} \bullet \mathbf{w}_n + \mathbf{v} \bullet \mathbf{w}_n)\mathbf{w}_n$.

(b) $\mathbf{u} \bullet \mathbf{v} = (\mathbf{u} \bullet \mathbf{w}_1)(\mathbf{v} \bullet \mathbf{w}_1) + \cdots + (\mathbf{u} \bullet \mathbf{w}_n)(\mathbf{v} \bullet \mathbf{w}_n)$. (This result is known as *Parseval's identity*.)